

Theoretical Assignment – Question 5

Schema, Fact & Dimension Tables, Star & Snowflake Schema, Database/Warehouse/Mart, OLTP & OLAP

1. What is a Schema?

A schema is the logical structure or blueprint of data. It describes what tables exist, what columns they contain, their data types, and how tables are related.

```
Customer  
■■■■■■■■■■  
customer_id  
customer_name  
city
```

```
Product  
■■■■■■■■■■  
product_id  
product_name  
category
```

```
Sales  
order_id  
customer_id  
product_id  
quantity  
amount
```

A schema describes the structure; it is different from the actual data stored in that structure.

Schema → How data is structured
Data → Actual values stored inside that structure

```
Example:
customer_id → Integer
customer_name → String
city → String
```

Actual row:
101 | Abhishek | Mysuru

Schema in a Data Warehouse

In a data warehouse, schema also describes how fact tables and dimension tables are organized and related.

Think of schema as the **blueprint of how data is organized**.

2. Fact Table vs Dimension Table

A useful way to understand them is: **Fact Table = What happened?** and **Dimension Table = Who, what, where, when, etc.?**

In a Retail business, a customer purchases a product from a store. The fact records the business event and measurements; dimensions provide descriptive context.

Fact Table

A Fact Table stores measurable business events or transactions.

[illegible]

```
date_id
quantity
unit_price
sales_amount
discount
```

Typical measures include quantity, sales_amount, discount, profit and cost. The table usually also contains foreign keys connecting it to dimensions.

Dimension Table

A Dimension Table contains descriptive information that provides context for facts.

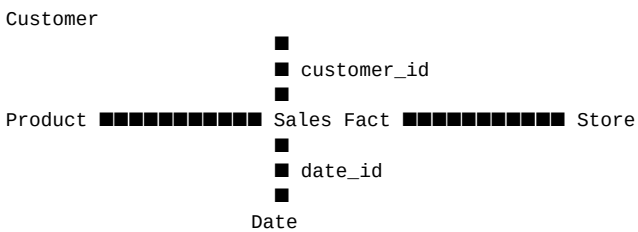
[illegible]

```
Product Dimension  
■■■■■■■■■■■■■■■■■■■■  
product_id  
product_name  
category  
brand
```

[illegible]

```
Date Dimension  
■■■■■■■■■■■■■■■■■■■■  
date_id  
date  
month  
year
```

How Fact and Dimension Tables Work Together



The Sales Fact tells us **what happened**, for example that two laptops were sold for ■1000. The dimensions tell us the context: who bought it, what product it was, where it was sold, and when.

Key Difference

Fact Table	Dimension Table
Stores business events	Stores descriptive information
Contains measurements	Contains attributes
Usually large	Usually smaller
Example: sales_amount	Example: customer_name
Used heavily for aggregation	Used for filtering/grouping
Contains foreign keys to dimensions	Usually contains keys referenced by facts

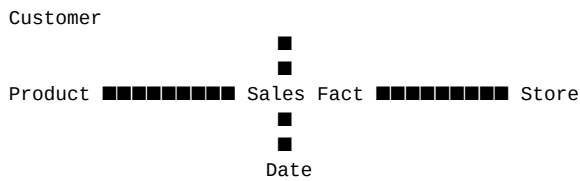
Memory trick: FACT → What happened? DIMENSION → Describe the event.

3. Star Schema vs Snowflake Schema

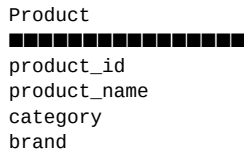
Both are data warehouse schema designs. The main difference is **how dimension tables are organized**.

Star Schema

In a **Star Schema**, one central fact table is directly connected to several dimension tables.

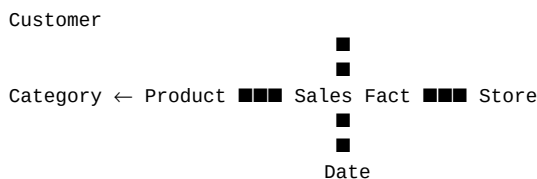


It is called a Star Schema because the structure resembles a star. Dimensions are generally more denormalized, so related descriptive information can remain together.



Snowflake Schema

A **Snowflake Schema** further normalizes dimension tables into additional related tables.



Instead of keeping all product information in one dimension, parts of the dimension can be separated into related tables such as Category and Brand.

Star vs Snowflake

Feature	Star Schema	Snowflake Schema
Dimension structure	More denormalized	More normalized
Number of tables	Generally fewer	Generally more
Joins	Fewer joins	More joins
Query structure	Simpler	More complex
Data redundancy	Higher	Lower
Design	Simpler	More complex

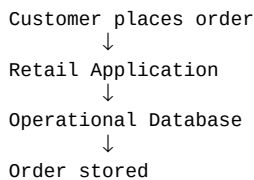
Fundamental difference: Star keeps dimensions relatively denormalized and directly connected to the fact table; Snowflake further normalizes dimensions into related tables.

4. Database vs Data Warehouse vs Data Mart

These systems can all store data, but they serve different purposes and scopes.

Database

A **database** is a system used to store and manage data for applications and business operations.



An operational database commonly handles day-to-day INSERT, UPDATE, DELETE and SELECT operations. Examples include MySQL, PostgreSQL, Oracle Database and SQL Server.

Data Warehouse

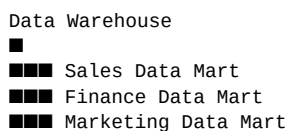
A **Data Warehouse** is a centralized system designed primarily for analytics and reporting. Data may be collected from multiple operational and external sources.



A warehouse commonly contains integrated and historical data used for questions such as total sales by year, product performance and regional trends.

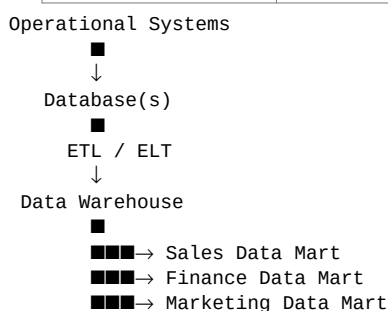
Data Mart

A **Data Mart** is a smaller, subject-specific portion of data used for a particular business area or department.



Database vs Data Warehouse vs Data Mart

Feature	Database	Data Warehouse	Data Mart
Main purpose	Application operations	Enterprise analytics	Department/subject analytics
Data	Operational/current focus	Historical + integrated	Focused subset
Scope	Application/business operation	Organization-wide	Specific department/subject
Typical workload	Transactions	Analytics/reporting	Analytics/reporting

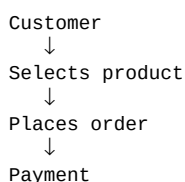


5. OLTP vs OLAP

A simple way to remember the difference is: **OLTP = running the business** and **OLAP = analyzing the business**.

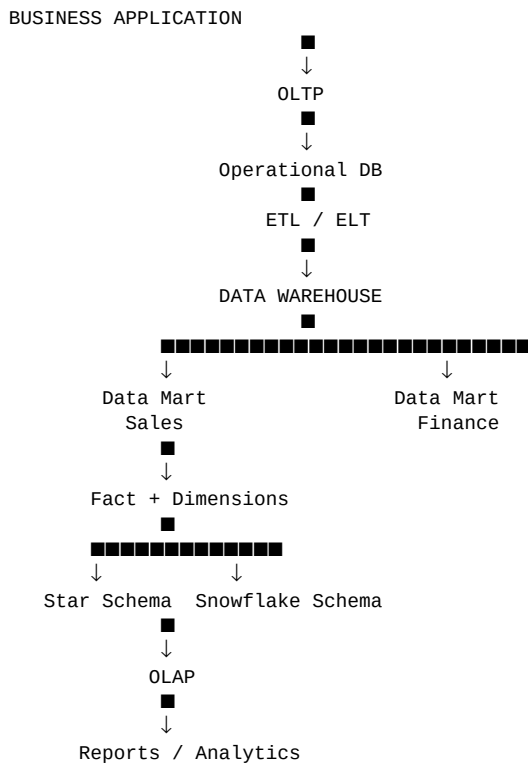
OLTP — Online Transaction Processing

OLTP systems are designed to handle **day-to-day transactions**.



OLTP → Capture the transaction.
OLAP → Analyze the transactions.

Complete Concept Connection



Quick Revision

Concept	Think of it as
Schema	Blueprint / structure of data
Fact Table	What happened / measurable events
Dimension Table	Context that describes the event
Star Schema	Fact in center + direct dimensions
Snowflake Schema	Fact + normalized dimensions
Database	Stores/serves operational application data
Data Warehouse	Centralized data for organization-wide analytics
Data Mart	Analytics data for a specific subject/department
OLTP	Run day-to-day transactions
OLAP	Analyze business data